

PAWTECT: A COMMUNITY-BASED WIRELESS DOG DETECTION AND ALERT SYSTEM FOR STRAY AND MISSING DOGS USING COMPUTER VISION AND RASPBERRY PI

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ABSTRACT

PawTect is a community-based wireless dog detection and alert system designed to mitigate public health risks and animal welfare issues in the Philippines, where approximately 12 million stray dogs and significant rabies fatalities persist. The system utilizes a YOLOv8-based computer vision model trained on diverse datasets, such as COCO and Stanford Dogs, to perform real-time species detection and breed classification. The hardware architecture employs a Raspberry Pi Zero as a decentralized edge computing node; this node performs initial detection and image compression at the source to minimize bandwidth consumption. These processed results are transmitted via LoRa wireless technology to a Raspberry Pi 4 central hub. This hub cross-references detected biometric features—such as facial landmarks and coat patterns—against a centralized SQL database containing registered owner profiles and pet characteristics. Upon a successful match, a API triggers immediate SMS notifications to owners or local authorities, ensuring a rapid response even in areas with unstable internet connectivity. This edge-to-hub integration provides a scalable, cost-effective framework for enhancing local government dog registries and promoting responsible pet ownership.

Keywords: Lora Wireless Transmission Systems, Real-Time Sms Notification, Dog Registration Program, Computer Vision Model, Raspberry Pi

INTRODUCTION

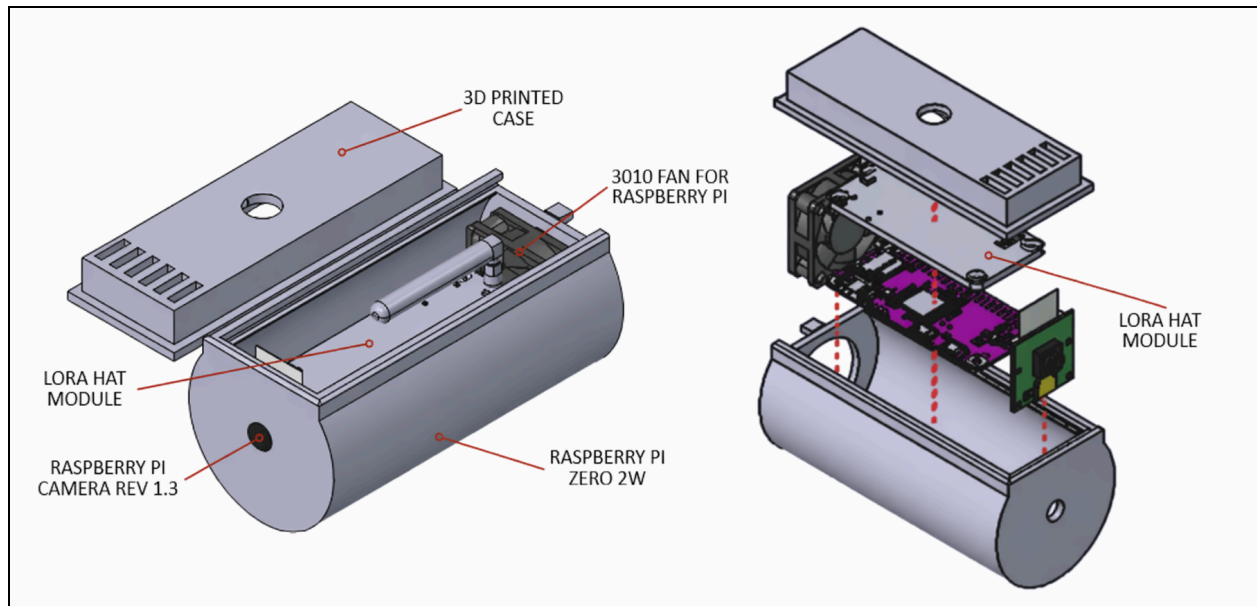


Figure 1. Computer-aided Drafting Design

In the Philippines, dogs are considered beloved companions in many households, serving as loyal pets, guardians, and family members. However, the growing population of stray and free-roaming dogs poses a serious threat to public health, community safety, and animal welfare.

Estimates from organizations such as the Philippine Animal Welfare Society (PAWS) indicate that there are approximately 12 million stray dogs nationwide, a figure frequently cited from 2019 to 2024 and still recognized as a major concern in 2025.

This large stray population significantly contributes to public health risks, particularly rabies, a viral disease that is almost always fatal once symptoms appear. In 2024, the Department of Health (DOH) reported 426 confirmed rabies cases, all resulting in death. About 45% of these cases were linked to bites from domestic pets, while others involved stray or unvaccinated animals. Although early 2025 data show a decline, with 55 cases recorded from January to March (a 39% decrease compared to the same period in 2024), rabies remains endemic and continues to claim hundreds of lives annually, especially among children.

Despite the implementation of Republic Act No. 9482, or the Anti-Rabies Act of 2007, which mandates annual dog registration, vaccination, and responsible ownership, compliance and enforcement remain inconsistent. Local government units (LGUs) are tasked with maintaining registries, conducting immunization programs, and managing stray populations, but progress is often hindered by limited resources, low public awareness, and economic challenges. Some areas, such as Manila, have introduced stricter measures like Ordinance No. 8587 (2020), requiring microchipping, registration, identification tags, and centralized databases, along with penalties for non-compliance.

Various programs and studies have attempted to address this issue through mass vaccination campaigns, community education, Catch-Neuter-Vaccinate-Return (CNVR) initiatives, and centralized databases for better tracking and pet recovery. Additionally, emerging technologies such as computer vision for animal recognition, Raspberry Pi-based systems, long-range communication through LoRa, and real-time SMS alerts have shown promise in both local and international contexts. However, these technologies are often implemented separately rather than as part of a unified, community-focused system suitable for the Philippine setting, where internet connectivity may be unreliable and manual reporting causes delays.

A major gap remains in the absence of an accessible, real-time, and integrated technological platform that combines accurate dog detection, owner registration databases, reliable wireless communication, and immediate alert systems. Existing approaches still rely heavily on manual processes, resulting in slower response times and inefficiencies in environments where dogs move freely and unpredictably.

To address this gap, this research proposes the development of PawTect, a community-based wireless dog detection and alert system. PawTect integrates computer vision with Raspberry Pi for automatic detection and classification of dogs by color and pattern, along with a centralized registration system for pet and owner information. It utilizes LoRa for long-range data transmission and a GSM module for real-time SMS notifications. The system aims to enhance community safety, promote responsible pet ownership, support LGUs in enforcing animal welfare laws, and contribute to reducing rabies cases. The study focuses on system design, development, model training, hardware-software integration, and initial performance testing within a controlled community, excluding nationwide deployment and long-term system maintenance.

Our system aligns with three Sustainable Development Goals (SDGs): Sustainable Development Goals 3, 11, and 15. Initially, SDG 3 Good Health and Well-being is vital since it

aids in illness prevention and guarantees public safety. The World Health Organization (2021) asserts that regulating dog populations and enhancing owner accountability can diminish rabies incidence globally. It also references SDG 11: Sustainable Cities and Communities, which aims to create inclusive, safe, resilient, and sustainable urban areas and human settlements. According to UN-Habitat (2020), the presence of stray dogs in communities affects safety, sanitation, and livability, which can be enhanced by data-driven animal management. This project corresponds with SDG 15: Life on Land, focusing on the ethical treatment of animals to safeguard biodiversity. The Food and Agriculture Organization (2019) asserts that monitoring mechanisms will prevent animal cruelty and mitigate the environmental consequences of uncontrolled dog populations. The SDG priorities indicate that responsible pet ownership, community safety, and animal welfare are significant global concerns that can be addressed through contemporary technological interventions.

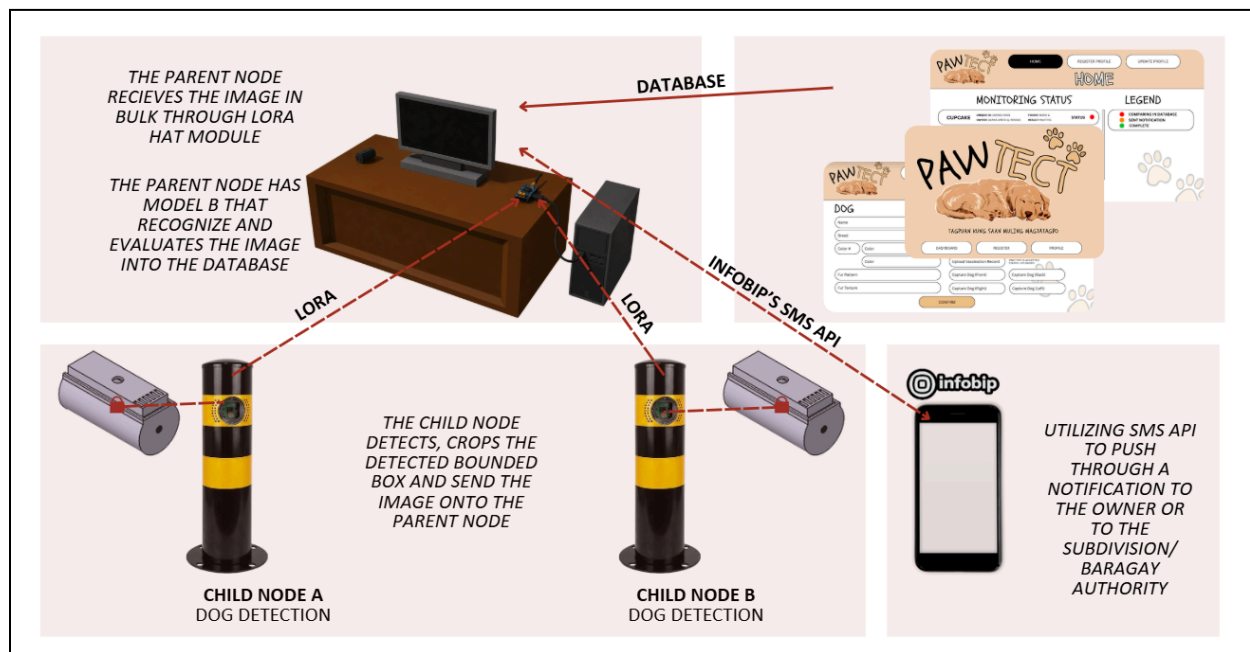


Figure 2. Conceptual Design

METHODOLOGY

The methodology employed in this study, detailing the systematic processes involved in the development, training, and implementation of a computer vision-based dog identification system with an integrated canine database and LoRa wireless transmission system for recognizing missing and stray dogs. To accomplish these objectives, the study is structured into multiple phases, each encompassing specific tasks and procedures that are built upon one another.

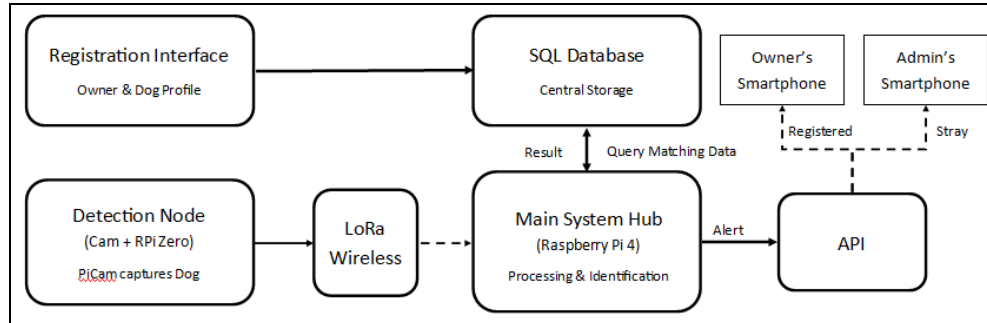


Figure 3. Conceptual Block Diagram

Phase I: Computer Vision Model Training

The first phase of the study focuses on the development and training of the computer vision model, which serves as the core of the PawTect system. In this phase, the researchers designed a model capable of detecting dogs from images and video streams and identifying their unique characteristics. The process involves training the model using multiple open-source datasets to ensure diversity in dog appearances, environments, and conditions. The system analyzes both facial and morphological features, such as landmarks, body shape, and fur patterns, to generate a comprehensive feature representation. These extracted features are then compared with stored data using similarity measures, allowing the system to predict the most probable matches and classify dogs based on breed and identity with corresponding confidence levels.

Phase II: Development of Dog Registration Application

The second phase of the study focuses on the development of the registration system, which is responsible for securely collecting owners' information and dog traits and then storing this data in the central database to facilitate future detection and verification operations. This phase expects input that includes the requirements gathered from the community and the database schema developed during initial planning. The expected output is a fully functional registration system containing (1) a dog registration application that accepts structured inputs, (2) a fully developed SQL database for storing owner and dog profiles, and (3) a user-friendly interface enabling users and administrators to effectively manage records. This subsystem establishes the basis for precise identification matching during close-range verification in the following phases.

Phase III: Building the Video Stream System

This phase focuses on building the video streaming system, which allows for real-time environmental monitoring, intended for the transmission of images across extensive distances utilizing LoRa Wireless Technology integrated with Raspberry Pi Zero and Cam. This advance aims to enhance computer vision capabilities, facilitating area monitoring, specifically within a community context, for the purposes of surveilling and locating stray or lost domesticated dogs.

Phase IV: Building the Main System

This phase focuses on constructing the central processing hub, or the brain of the entire operation using a Raspberry Pi 4. This central unit is responsible for receiving data from subsystems, processing information, querying databases, and issuing alerts. The entire phase is conducted in a laboratory or workshop setting, and the respondents of the study are the project researchers and developers themselves, who perform the integration and testing tasks.

Phase V: Troubleshooting and System Evaluation

The fifth and last phase of the study focuses on the assessment of the overall system's performance, accuracy, and reliability following the integration of all primary components. The input for this phase includes the fully functional subsystems, the video stream subsystem (Raspberry Pi Zero), the main communication and decision hub (Raspberry Pi 4), the computer vision model, the LoRa communication modules, and the use of API alert system. The expected 56 outputs are a validated and optimized system, including results on dog recognition accuracy, close-range verification efficacy, communication latency, camera reliability, and SMS alert responsiveness. This phase ensures that the system is ready for deployment and can function reliably in real-world field conditions.

RESULTS AND DISCUSSION

This chapter presents the findings derived from the data collection process, followed by a comprehensive analysis and interpretation. The results are organized according to the specific research objectives outlined in the first chapter. By synthesizing the empirical evidence with established theoretical frameworks, this section aims to clarify the implications of the findings and address the central research questions of this study.

Components	Description	Function in the System
PC	A general-purpose computing device	Serves as the main processing hub and is used for system monitoring, data storage, and user interface.
Raspberry Pi Zero 2W	A compact single-board computer with built-in Wi-Fi and Bluetooth	It performs edge processing; receives input from the camera, detects/captures dog data, and transmits it to the main hub.
Raspberry Pi Camera Rev 1.3	A camera module compatible with Raspberry Pi boards	Captures images of detected dogs for processing
LoRa Hat	A long-range wireless communication module attached to the Raspberry Pi	Enables long-distance data transmission between devices

Table 1. Main Components

COMPUTER VISION MODEL TRAINING PERFORMANCE AND TRIALS

Metric	Result
Dataset Used	COCO – 5,000 Dog Images Google Images – 94 Images
Total Trials	30
Total Images Evaluated	300
True Positives (TP)	268
False Positives (FP)	21
False Negatives (FN)	32
True Negatives (TN)	279
Accuracy	90.2%
Precision	92.7%
Recall	89.3%
F1-Score	91.0%

Table 2. Confusion Matrix for Model A – Dog Detection Performance

Model A was deployed on the Raspberry Pi Zero 2W and is responsible for detecting dogs from input images captured using the Raspberry Pi camera module. The model was trained using a combination of approximately 5,000 dog images from the COCO dataset and 94 additional images collected from Google Images, ensuring diversity in dog breeds, poses, and environmental conditions.

Across 30 evaluation trials, which involved a total of 300 image inputs, Model A achieved an overall accuracy of 90.2%, with a precision of 92.7%, a recall of 89.3%, and an F1-score of 91.0%. These results indicate that the model performs reliably in distinguishing dogs from non-dog objects in various environments.

The high precision value suggests that the model is highly effective in minimizing false detections, while the recall value indicates that most dog instances are successfully identified. However, some false negatives were still observed, primarily due to environmental factors such as low lighting, motion blur, and partial occlusion of subjects.

Despite operating on a resource-constrained device, the Raspberry Pi Zero 2W was able to successfully execute the detection model, demonstrating the feasibility of edge computing for real-time object detection. The output of Model A is then used to crop and extract detected dog regions, which are forwarded to Model B for further processing.

Metric	Result
Dataset Used	15 Registered Dogs
Total Trials	30
Input Images (from Model A)	268
Correct Identifications	231
Misclassifications	37
Recognition Accuracy	86.2%
Precision	88.8%
Recall	86.2%
F1-Score	87.5%

Table 3. Confusion Matrix for Model B – Dog Recognition Performance

Model B is executed on a personal computer and is responsible for identifying individual dogs based on the cropped outputs received from Model A. This model was trained using a controlled dataset consisting of 15 registered dogs from 10 different owners, where each dog was captured from multiple perspectives, including front, back, left, and right views.

In the same 30 evaluation trials, Model B processed a total of 268 cropped dog images generated by Model A. From these inputs, the system correctly identified 231 instances, while 37 cases were misclassified, resulting in a recognition accuracy of 86.2%. Additionally, the model achieved a precision of 88.8%, a recall of 86.2%, and an F1-score of 87.5%.

The PC-based deployment allowed for more stable and efficient processing compared to edge execution, enabling more accurate feature extraction and comparison. However, the performance of Model B is limited by the relatively small dataset size. Since only 10 registered dogs were used, the model has limited exposure to variations in appearance, which affects its ability to generalize under different conditions.

LORA TRANSMISSION PERFORMANCE AND TRIALS

The LoRa module used in this project is Waveshare SX1262 LoRa HAT for Raspberry Pi capable of 850.125~930.125MHz frequency band, its speed can reach 62.5 kbps according to its specs in the website and receive sensitivity of -147dBm, the power consumption of this device is 133 mA and sleep current is 2 μ A, its communication interface is UART (TTL level) which is not as powerful as SPI.

The program developed to test this module is simple it sends data at fixed rate of < 100 bytes a chunk which is sent every 0.5 seconds, reaching going lesser than 0.5 seconds causes the module to not handle the data properly causing loss in data, the distance between the nodes being tested are 5 meters, these are the throughput for each file size transferred using this simple program.

File Size	Chunks	Duration to Completion
10KB	103	62.5 seconds
20KB	205	102.5 seconds
30KB	308	154 seconds
40KB	410	205 seconds
50KB	512	256 seconds

The observed pattern is a consistent ratio of 10 chunks per kilobyte (KB), which logically translates to seconds. For instance, 50KB equates to approximately 512 chunks, when halved, it approximates to 256 seconds. This phenomenon is likely attributed to the device's enhanced sensitivity to data loss at extended distances, thereby necessitating a more frequent chunking process to ensure data integrity at long distances.

SUMMARY AND CONCLUSION

This section presents a summary of the study's key findings and concludes with the overall insights derived from the results.

This study successfully developed a system for detecting and recognizing dogs by integrating edge computing, computer vision, and long-range communication technologies. The system is composed of a PC, Raspberry Pi Zero 2W, Raspberry Pi Camera, and a LoRa HAT module, each contributing to efficient system operation. The Raspberry Pi Zero 2W performs real-time edge processing by capturing and detecting dogs from images, while the PC serves as the central hub for recognition, monitoring, and data storage. The LoRa module enables reliable long-distance communication between system components.

The system's performance was evaluated using two models. Model A, which is responsible for dog detection, was trained using a diverse dataset consisting of 5,000 images from the COCO dataset and 94 images from Google Images. Across 30 trials and 300 test images, it achieved an accuracy of 90.2%, precision of 92.7%, recall of 89.3%, and an F1-score of 91.0%. These results demonstrate that the model is highly effective in accurately detecting dogs across various conditions.

Model B, responsible for dog recognition, was trained using a dataset of 15 registered dogs. It processed 268 cropped images from Model A and achieved an accuracy of 86.2%, precision of 88.8%, recall of 86.2%, and an F1-score of 87.5%. The results indicate that the model is capable of reliably identifying individual dogs, showing strong performance given the controlled dataset.

The LoRa transmission tests showed consistent and stable performance in data transfer. The system successfully transmitted data in small chunks at fixed intervals, maintaining reliability across different file sizes. The observed transmission pattern demonstrates the suitability of LoRa for dependable long-range communication within the system.

The results of the study confirm that the proposed system effectively meets its objective of detecting and recognizing dogs using a combination of edge computing and centralized processing. Model A demonstrated high accuracy and strong reliability in detecting dogs, even when deployed on a resource-constrained device, highlighting the effectiveness of edge-based processing for real-time applications.

Model B also achieved strong recognition performance, successfully identifying individual dogs with a high level of accuracy. This indicates that the system is capable of performing detailed analysis and classification when supported by a central processing unit.

Additionally, the integration of the LoRa module proved to be a reliable solution for long-distance communication, enabling consistent data transmission between system components. The stable transmission performance further supports the practicality of the system in real-world applications.

Overall, the study demonstrates that the integration of computer vision, edge computing, and LoRa communication is an effective and feasible approach for developing an automated dog detection and recognition system. The findings contribute valuable insights into the application of these technologies and provide a strong foundation for future enhancements and broader implementations.